

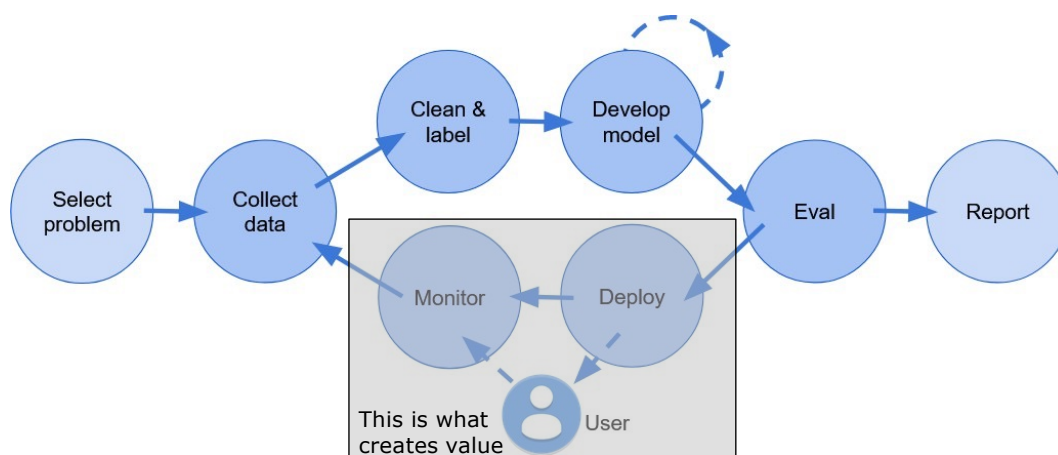
Data Distribution Shifts & Monitoring



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Remember this figure

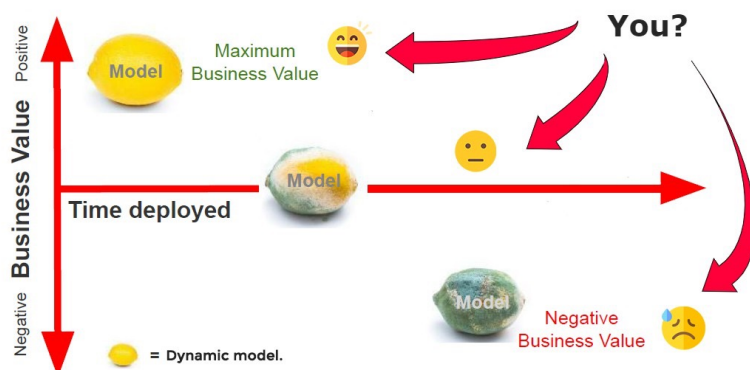


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We are in the endgame now

Machine Learning models are dynamic and degrade over time after being deployed to production.



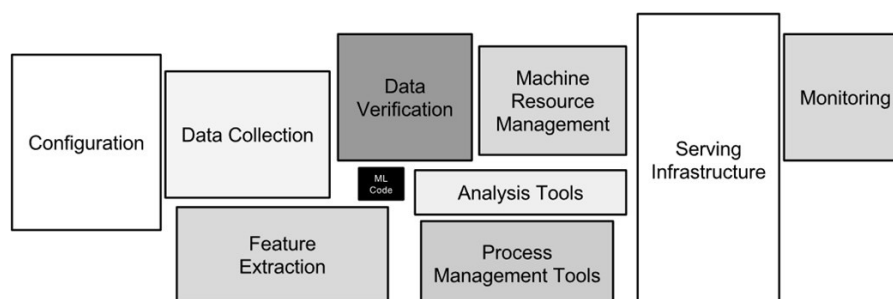
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Operations is hard

"All models are wrong, but some are useful" - George Box

"87% of data science projects never make it into production"



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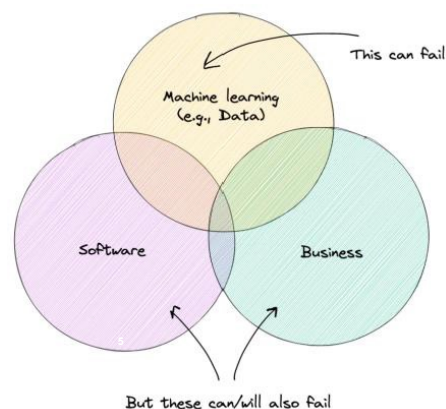
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What can fail?

▶ Monitoring deals with things ~~if~~ when they break

Applications will fail for many reasons, but we can group into three

- ML failures
- Software failures
- Business failures



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Software failures

"Software is never done (only abandoned)"

All the reasons a non-ML application already can fail

- 💡 Dependencies
- 💡 Deployments
- 💡 Hardware
- 💡 Downtime/crashing



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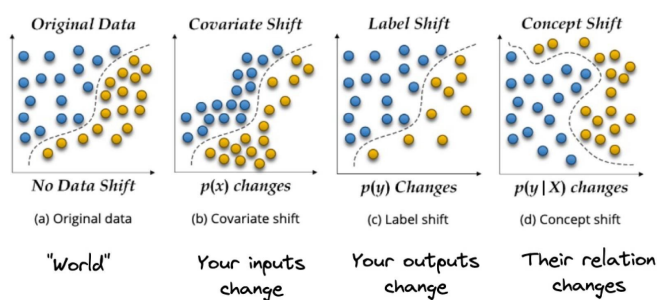
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ML failures

ML specific applications can fail for even more reasons

- 💡 Edge cases
- 💡 Feedback loops
- 💡 Training != production

x : your inputs
 y : your outputs



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ML failures

- ⚠️ Data drift
 Model perform worse on unseen data
- ⚠️ Target drift
 The world have changed, you need to wake up

Deal with by

- ✅ Train model on massive dataset initially
- ✅ Domain adaptation of large models
- ✅ Retrain from scratch or finetune



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ML failures

To know how you are doing, you most likely is going to need labels for incoming data.

Three ways to get them:

- 🔥 Hand labels
Annotate by hand, expensive as hell
- 🔥 Natural labels
You get correct label in the future, leverage it
- 🔥 Programmatic labels
Any feedback is better than non, get creative



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9

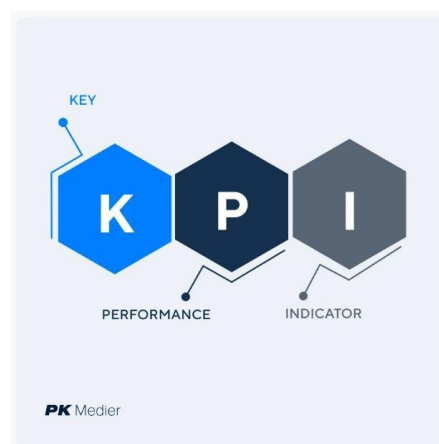
Business failures

When ML deployments are not synchronized with business, then it fails

- 💡 We are not doing enough predictions
- 💡 Model predictions are not benefitting KPI

Deal with by

- ✅ Better business alignment



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Logs

💡 Logs are textual or structured records generated by applications

💡 They provide a detailed account of events, errors, warnings, and informational messages that occur during the operation of the system

💡 Logs are essential for diagnosing issues, debugging, and auditing.

Traces

💡 Traces are detailed records of specific transactions or events as they move through a system.

💡 A trace typically includes information about the sequence of operations, timing, and dependencies between different components.

💡 Traces help in understanding the flow of a request or a transaction across different components.

Metrics

💡 Quantitative measurements of the system.

💡 They are usually numbers that are aggregated over a period of time. E.g. the number of requests per minute.

💡 Metrics are used to get an overview of the system



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11

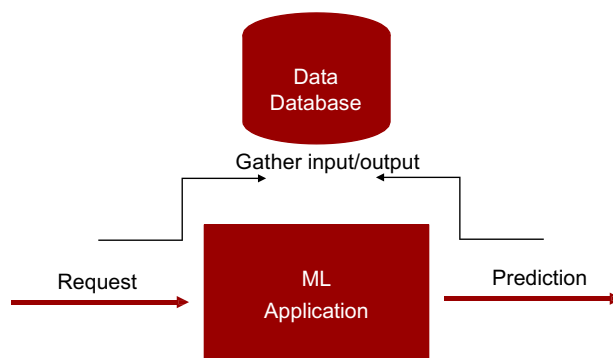
Let's setup a system



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12

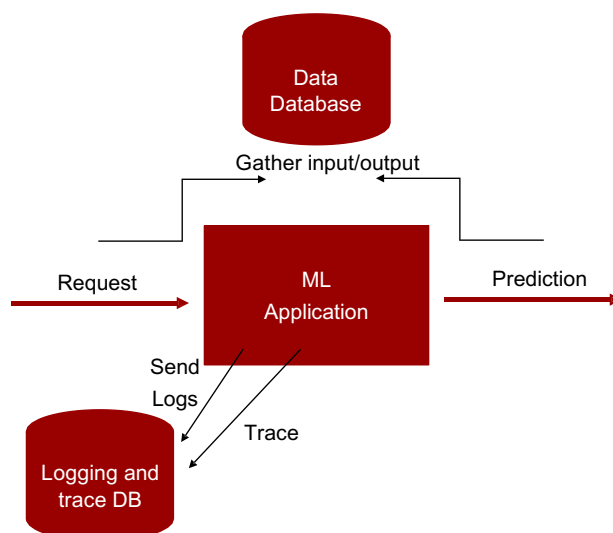
Let's setup a system



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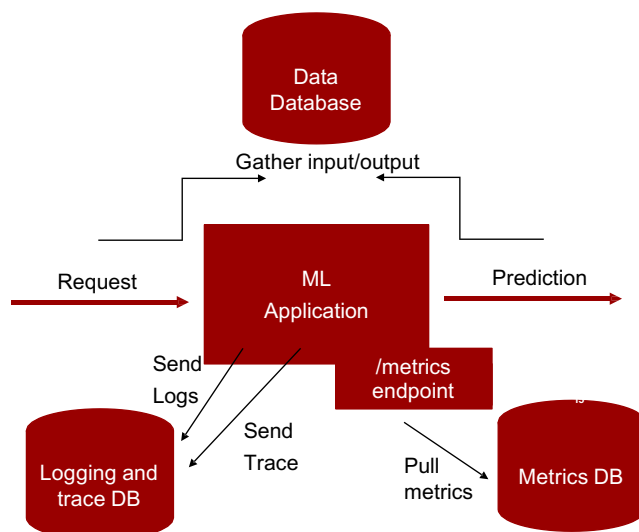
Let's setup a system



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14

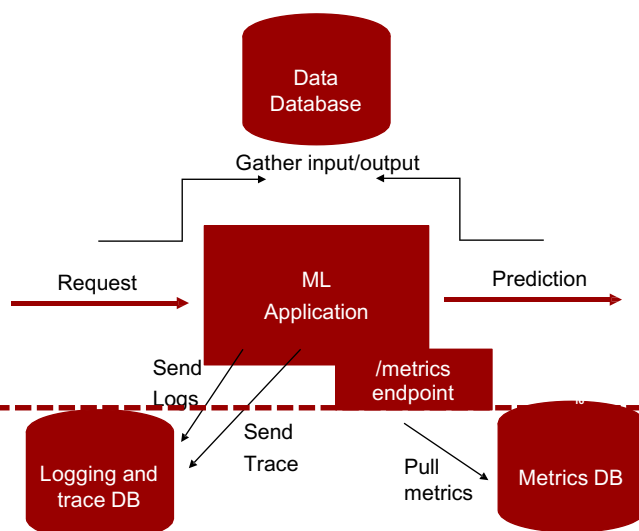
Let's setup a system



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15

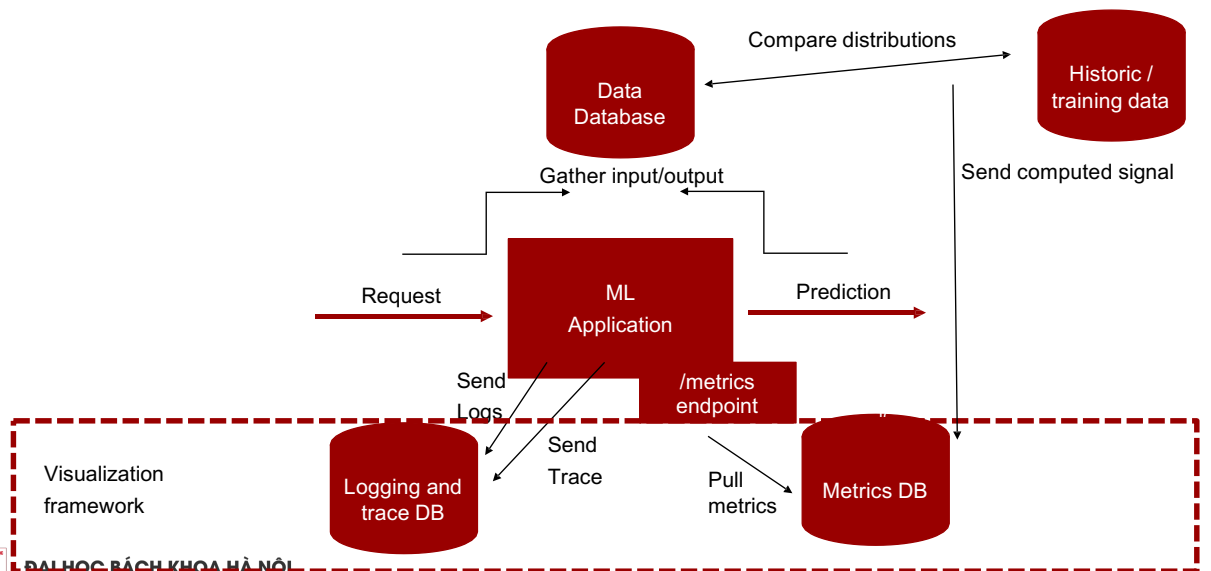
Let's setup a system



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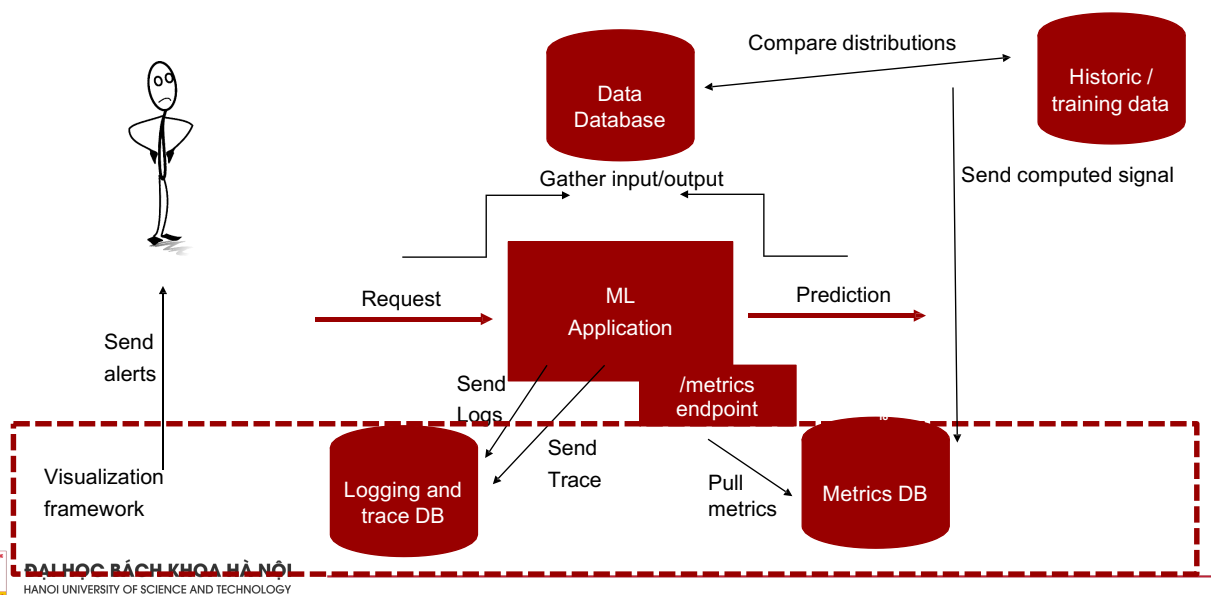
16

Let's setup a system



17

Let's setup a system



18

Natural labels & feedback loops



19

Natural labels

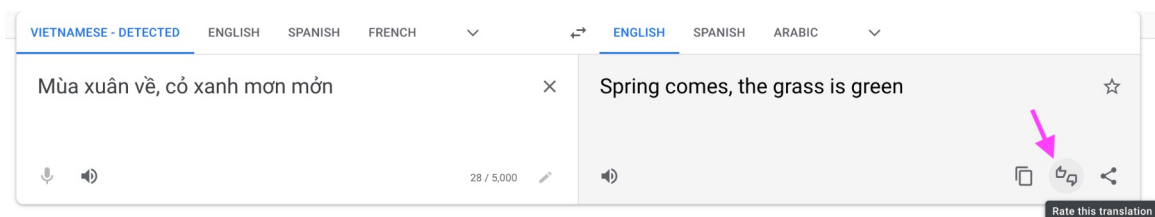
- The model's predictions can be automatically evaluated or partially evaluated by the system.
- Examples:
 - ETA
 - Ride demand prediction
 - Stock price prediction
 - Ads CTR
 - Recommender system



20

Natural labels

- You can engineer a task to have natural labels

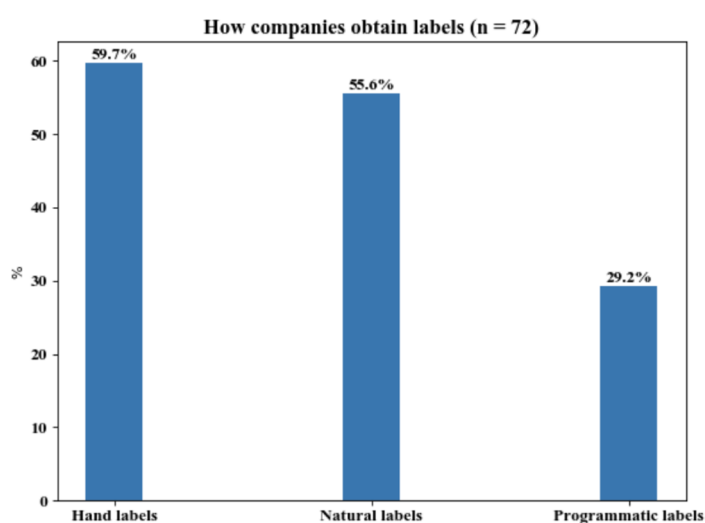


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21

21

Natural labels: surprisingly common



⚠ Biases ⚠

- Small sample size
- Companies might only use ML for tasks with natural labels

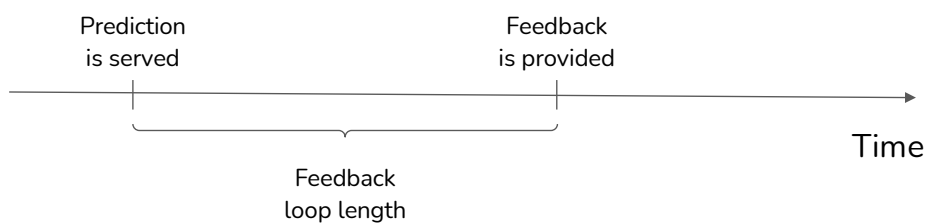


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[Claypot AI's real-time ML survey \(2022\)](#)

22

22

Delayed labels

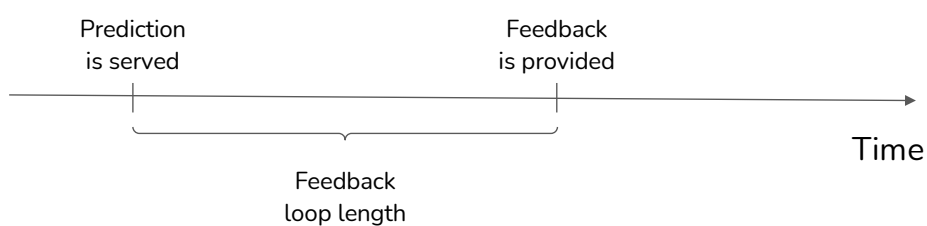


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23

Delayed labels



- Short feedback loop: minutes -> hours
 - Reddit / Twitter / TikTok's recommender systems
- Long feedback loop: weeks -> months
 - Stitch Fix's recommender systems
 - Fraud detection

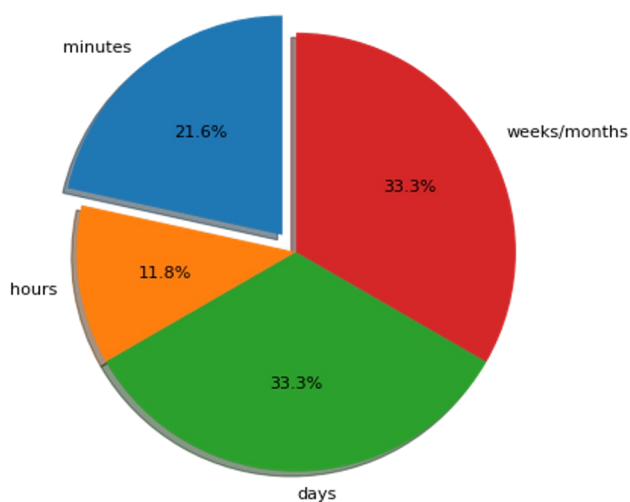


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24

24

Feedback loop length (n = 51)



25

⚠ Labels are often assumed ⚠

- Recommendation:
 - Click -> good rec
 - After X minutes, no click -> bad rec
 - Too short
False negatives
 - Too long
Slow feedback

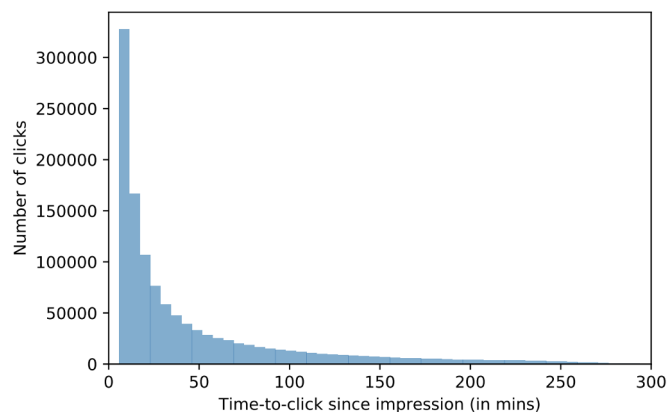
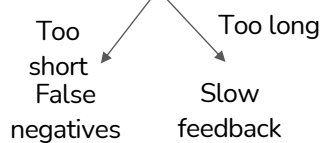
Speed vs. accuracy
tradeoff

26

⚠ Labels are often assumed ⚠

- Recommendation:

- Click -> good rec
- After X minutes, no click -> bad rec



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Addressing Delayed Feedback for Continuous Training with Neural Networks in CTR prediction (Ktena et al., 2019)

27

27

Causes of ML failures



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28

28

Amazon scraps secret AI recruiting tool that showed bias against women

That is because Amazon's computer models were trained to vet applicants by observing patterns in resumes submitted to the company over a 10-year period. Most came from men, a reflection of male dominance across the tech industry.

In effect, Amazon's system taught itself that male candidates were preferable. It penalized resumes that included the word "women's," as in "women's chess club captain." And it downgraded graduates of two all-women's colleges, according to people familiar with the matter. They did not specify the names of the schools.



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[Amazon scraps secret AI recruiting tool that showed bias against women](#) (Reuters, 2018)

29

29

Japan's Henn na Hotel fires half its robot workforce

"Guests complained their robot room assistants thought snoring sounds were commands and would wake them up repeatedly during the night."



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<https://www.hotelmanagement.net/tech/japan-s-henn-na-hotel-fires-half-its-robot-workforce>

30

30

What is an ML failure?

A failure happens when one or more expectations of the system is violated.

Two types of expectations:

- Operational metrics: e.g. average latency, throughput, uptime
- ML metrics: e.g. accuracy, F1, BLEU score



What is an ML failure?

A failure happens when one or more expectations of the system is violated

- Traditional software: mostly operational metrics
- ML systems: operational + ML metrics
 - Ops: returns an English translation within 100ms latency on average
 - ML: BLEU score of 55 (out of 100)



ML system failures

- If you enter a sentence and get no translation back -> ops failure
- If one translation is incorrect -> ML failure?



33

ML system failures

- If you enter a sentence and get no translation back -> ops failure
- If one translation is incorrect -> ML failure?

- Not necessarily: expected BLEU score < 100
- ML failure if translations are consistently incorrect

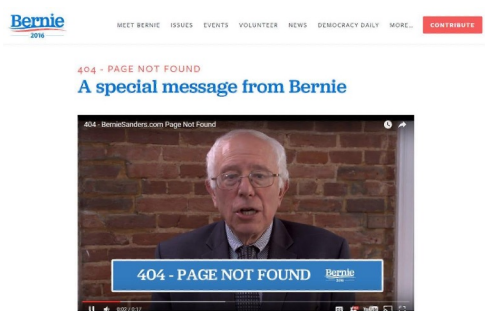


34

Ops failures

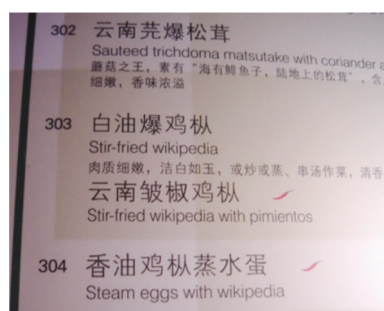
Visible

- 404, timeout, segfault, OOM, etc.



ML failures

Often invisible



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Causes of ops failures (software system failures)

- Dependency failures
- Deployment failures
- Hardware failures
- Network failure: downtime / crash



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36

36

Causes of ops failures (software system failures)

- Dependency failures
- Deployment failures
- Hardware failures
- Network failure: downtime / crash



60 / 96 ML systems failures are non-ML failures

(Papasian & Underwood, 2020)

As tooling & best practices around ML production mature,
there will be less surface for software failures



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ML-specific failures (during/post deployment)

1. Production data differing from training data
2. Edge cases
3. Degenerate feedback loops

We've already covered problems pre-
deployment in previous lectures!



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38

38

Production data differing from training data

- Train-serving skew:
 - Model performing well during development but poorly after production
- Data distribution shifts
 - Model performing well when first deployed, but poorly over time
 - ⚠️ What looks like data shifts might be caused by human errors ⚠️



Production data differing from training data

- Train-serving skew:
 - Model performing well during development but poorly after production
 - Data distribution shifts
 - Model performing well when first deployed, but poorly over time
 - ⚠️ What looks like data shifts might be caused by human errors ⚠️
- } Common & crucial.
Will go into detail!



Edge cases

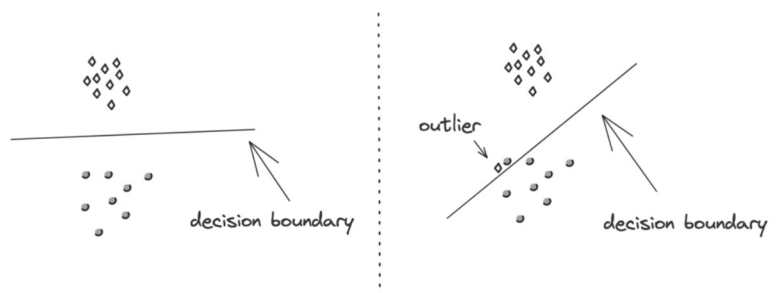
- Self-driving car (yearly)
 - Safely: 99.99%
 - Fatal accidents: 0.01%



41

Edge case vs. outlier

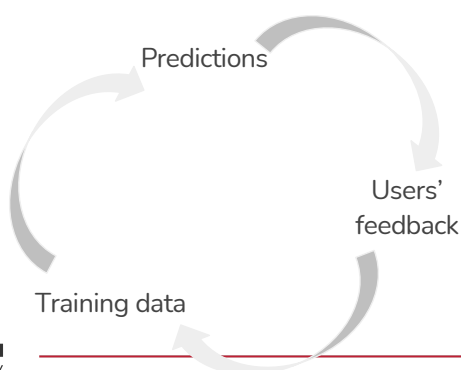
- Outliers
 - Refer to inputs
 - Options to ignore/remove
- Edge cases
 - Refer to outputs
 - Can't ignore/remove



42

Degenerate feedback loops

- When predictions influence the feedback, which is then used to extract labels to train the next iteration of the model
- Common in tasks with natural labels



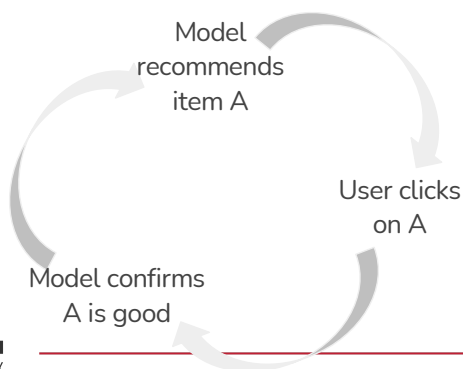
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43

43

Degenerate feedback loops: recsys

- Originally, A is ranked marginally higher than B -> model recommends A
- After a while, A is ranked much higher than B



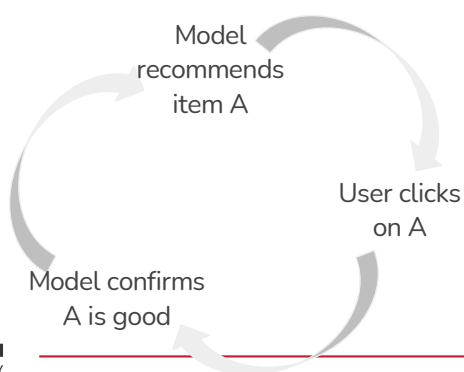
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44

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Degenerate feedback loops: recsys

- Originally, A is ranked marginally higher than B -> model recommends A
- After a while, A is ranked much higher than B



Over time,
recommendations
become more
homogenous



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45

Degenerate feedback loops: resume screening

- Originally, model thinks X is a good feature
- Model only picks resumes with X
- Hiring managers only see resumes with X, so only people with X are hired
- Model confirms that X is good

Replace X with:

- Has a name that is typically used for males
- Went to Stanford
- Worked at Google



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46

46

Degenerate feedback loops: resume screening

- Originally, model thinks X is a good feature
- Model only picks resumes with X
- Hiring managers only see resumes with X, so only people with X are hired
- Model confirms that X is good

Tracking feature importance might help!



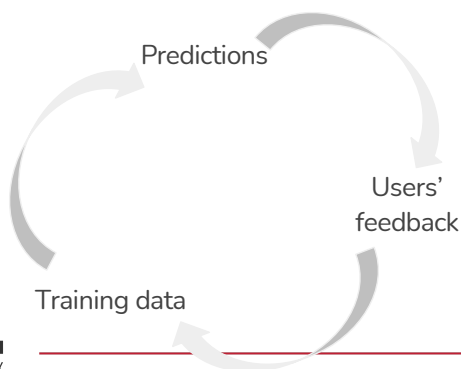
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47

47

Detecting degenerate feedback loops

Only arise once models are in production -> hard to detect during training



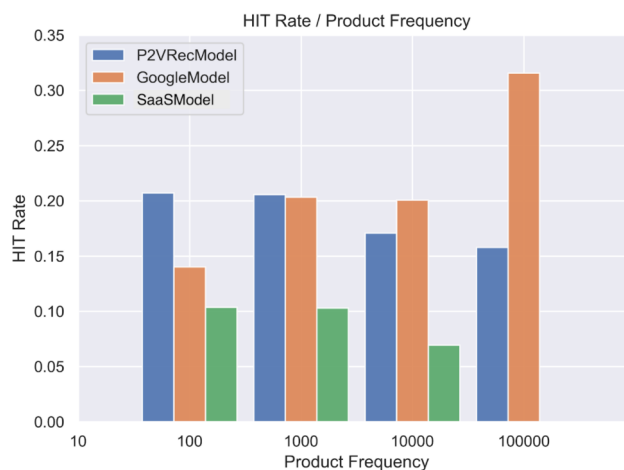
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48

48

Degenerate feedback loops: detect

- Average Rec Popularity (ARP)
 - Average popularity of the recommended items
- Average Percentage of Long Tail Items (APLT)
 - average % of long tail items being recommended
- Hit rate against popularity
 - Accuracy based on recommended items' popularity buckets



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[Beyond NDCG: behavioral testing of recommender systems with RecList](#) (Chia et al., 2021)

49

49

Degenerate feedback loops: mitigate

1. Randomization
2. Positional features



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50

50

Randomization

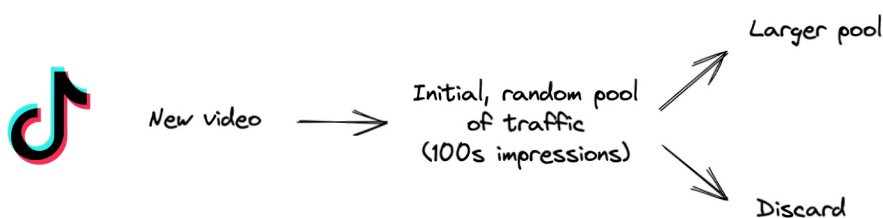
- Degenerate feedback loops increase output homogeneity
- Combat homogeneity by introducing randomness in predictions



51

Randomization

- Degenerate feedback loops increase output homogeneity
- Combat homogeneity by introducing randomness in predictions
- Recsys: show users random items & use feedback to determine items' quality



52

Positional features

- If a prediction's position affects its feedback in any way, encode it.
 - Numerical: e.g. position 1, 2, 3, ...
 - Boolean: e.g. shows first position or not



Positional features: naive

ID	Song	Genre	Year	Artist	User	1st Position	Click
1	Shallow	Pop	2020	Lady Gaga	listenr32	False	No
2	Good Vibe	Funk	2019	Funk Overlord	listenr32	False	No
3	Beat It	Rock	1989	Michael Jackson	fancypants	False	No
4	In Bloom	Rock	1991	Nirvana	fancypants	True	Yes
5	Shallow	Pop	2020	Lady Gaga	listenr32	True	Yes



Positional features: naive

ID	Song	Genre	Year	Artist	User	1st Position	Click
1	Shallow	Pop	2020	Lady Gaga	listenr32	False	No
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5	Shallow	Pop	2020	Lady Gaga	listenr32	True	Yes

Doesn't have this
feature during
inference?



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55

Positional features: naive

ID	Song	Genre	Year	Artist	User	1st Position	Click
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4	In Bloom	Rock	1991	Nirvana	fancypants	True	Yes
5	Shallow	Pop	2020	Lady Gaga	listenr32	True	Yes

Set to False during
inference



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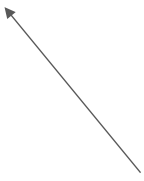
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56

Positional features: 2 models

1. Predicts the probability that the user will **see and consider** a recommendation given its position.
2. Predicts the probability that the user will **click on the item given that they saw and considered it.**

Model 2 doesn't
use positional
features




How might degenerate feedback loops occur? (10 mins)

1. Build a system to predict stock prices and use the predictions to make buy/sell decisions.
2. Use text scraped from the Internet to train a language model, then use the same language model to generate posts.

Discuss how you might mitigate the consequences of these feedback loops.



Data distribution shifts



59

- Source distribution: data the model is trained on
- Target distribution: data the model runs inference on



60

Supervised learning: $P(X, Y)$

1. $P(X, Y) = P(Y|X)P(X)$
2. $P(X, Y) = P(X|Y)P(Y)$



Types of data distribution shifts

Type	Meaning	Decomposition
Covariate shift	$P(X)$ changes $P(Y X)$ remains the same	$P(X, Y) = P(Y X)P(X)$
Label shift	$P(Y)$ changes $P(X Y)$ remains the same	$P(X, Y) = P(X Y)P(Y)$
Concept drift	$P(X)$ remains the same $P(Y X)$ changes	$P(X, Y) = P(Y X)P(X)$



Covariate shift

- $P(X)$ changes
- $P(Y|X)$ remains the same

- Statistics: a covariate is an independent variable that can influence the outcome of a given statistical trial.
- Supervised ML: input features are covariates



Covariate shift

- $P(X)$ changes
- $P(Y|X)$ remains the same

- Statistics: a covariate is an independent variable that can influence the outcome of a given statistical trial.
- Supervised ML: input features are covariates
- Input distribution changes, but for a *given input*, output is the same



Covariate shift: example

- Predicts $P(\text{cancer} \mid \text{patient})$
 - $P(\text{age} > 40)$: training > production
 - $P(\text{cancer} \mid \text{age} > 40)$: training = production
- $P(X)$ changes
 - $P(Y|X)$ remains the same



Covariate shift: causes (training)

- Data collection
 - E.g. women >40 are encouraged by doctors to get checkups
 - Closely related to sampling biases
 - Training techniques
 - E.g. oversampling of rare classes
 - Learning process
 - E.g. active learning
- Predicts $P(\text{cancer} \mid \text{patient})$
 - $P(\text{age} > 40)$:
 - training > production
 - $P(\text{cancer} \mid \text{age} > 40)$:
 - training = production
- $P(X)$ changes
 - $P(Y|X)$ remains the same



Covariate shift: causes (prod)

- $P(X)$ changes
- $P(Y|X)$ remains the same

Changes in environments

- Ex 1: $P(\text{convert to paid user} | \text{free user})$
 - New marketing campaign attracting users from with higher income
 - $P(\text{high income})$ increases
 - $P(\text{convert to paid user} | \text{high level})$ remains the same



Covariate shift: causes (prod)

- $P(X)$ changes
- $P(Y|X)$ remains the same

Changes in environments

- Ex 2: $P(\text{Covid} | \text{coughing sound})$
 - Training data from clinics, production data from phone recordings
 - $P(\text{coughing sound})$ changes
 - $P(\text{Covid} | \text{coughing sound})$ remains the same



Covariate shift

- Research: if knowing in advance how the production data will differ from training data, use [importance weighting](#)
- Production: unlikely to know how a distribution will change in advance



Label shift

- $P(Y)$ changes
 - $P(X|Y)$ remains the same
- Output distribution changes but for a *given output*, input distribution stays the same.



Label shift & covariate shift

- Predicts $P(\text{cancer} \mid \text{patient})$
- $P(\text{age} > 40)$: training > production
- $P(\text{cancer} \mid \text{age} > 40)$: training = production
- $P(\text{cancer})$: training > production
- $P(\text{age} > 40 \mid \text{cancer})$: training = prediction
- $P(X)$ changes
- $P(Y|X)$ remains the same
- $P(Y)$ changes
- $P(X|Y)$ remains the same

*$P(X)$ change often leads to $P(Y)$ change, so
covariate shift often means label shift*



Label shift & covariate shift

- Predicts $P(\text{cancer} \mid \text{patient})$
- New preventive drug: reducing $P(\text{cancer} \mid \text{patient})$ for all patients
- $P(\text{age} > 40)$: training > production
- $P(\text{cancer} \mid \text{age} > 40)$: training > production
- $P(\text{cancer})$: training > production
- $P(\text{age} > 40 \mid \text{cancer})$: training = prediction
- $P(X)$ changes
- ~~$P(Y|X)$ remains the same~~
- $P(Y)$ changes
- $P(X|Y)$ remains the same

Not all label shifts are covariate shifts!



Concept Drift

- $P(X)$ remains the same
- $P(Y|X)$ changes

- Same input, expecting different output
- $P(\text{houses in SF})$ remains the same
- Covid causes people to leave SF, housing prices drop
 - $P(\$5M | \text{houses in SF})$
 - Pre-covid: high
 - During-covid: low



Concept Drift

- $P(X)$ remains the same
- $P(Y|X)$ changes

- Concept drifts can be cyclic & seasonal
 - Ride sharing demands high during rush hours, low otherwise
 - Flight ticket prices high during holidays, low otherwise



General data changes

- Feature change
 - A feature is added/removed/updated



General data changes

- Feature change
 - A feature is added/removed/updated
- Label schema change
 - Original: {"POSITIVE": 0, "NEGATIVE": 1}
 - New: {"POSITIVE": 0, "NEGATIVE": 1, "NEUTRAL": 2}



Detecting data distribution shifts

How to determine that two distributions are different?



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77

77

Detecting data distribution shifts

How to determine that two distributions are different?

1. Compare statistics: mean, median, variance, quantiles, skewness, kurtosis, ...
 - Compute these stats during training and compare these stats in production



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78

78

Detecting data distribution shifts

How to determine that two distributions are different?

1. Compare statistics: mean, median, variance, quantiles, skewness, kurtosis, ...
 - Not universal: only useful for distributions where these statistics are meaningful



Detecting data distribution shifts

How to determine that two distributions are different?

1. Compare statistics: mean, median, variance, quantiles, skewness, kurtosis, ...
 - Not universal: only useful for distributions where these statistics are meaningful
 - Inconclusive: if statistics differ, distributions differ. If statistics are the same, distributions can still differ.



Cumulative vs. sliding metrics

- Sliding: reset at each new time window



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This image is based on an example from [MadeWithML](#) (Goku Mohandas).

81

81

Detecting data distribution shifts

How to determine that two distributions are different?

1. Compare statistics: mean, median, variance, quantiles, skewness, kurtosis, ...
2. Two-sample hypothesis test
 - Determine whether the difference between two populations is statistically significant
 - If yes, likely from two distinct distributions

E.g.

1. Data from yesterday
2. Data from today



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82

82

Two-sample test: KS test (Kolmogorov–Smirnov)

- Pros
 - Doesn't require any parameters of the underlying distribution
 - Doesn't make assumptions about distribution
- Cons
 - Only works with one-dimensional data

- ↙
- Useful for prediction & label distributions
 - Not so useful for features



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83

83

Two-sample test

Drift Detection

Detector	Tabular	Image	Time Series	Text	Categorical Features	Online	Feature Level
Kolmogorov-Smirnov	✓	✓		✓	✓		✓
Maximum Mean Discrepancy	✓	✓		✓	✓	✓	
Learned Kernel MMD	✓	✓		✓	✓		
Least-Squares Density Difference	✓	✓		✓	✓	✓	
Chi-Squared	✓				✓		✓
Mixed-type tabular data	✓				✓		✓
Classifier	✓	✓	✓	✓	✓		
Spot-the-diff	✓	✓	✓	✓	✓		✓
Classifier Uncertainty	✓	✓	✓	✓	✓		
Regressor Uncertainty	✓	✓	✓	✓	✓		



[alibi-detect](#) (OS)

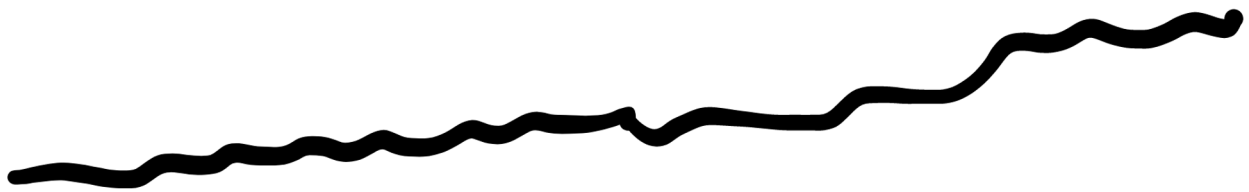
Most tests work better on low-dim data, so dim reduction is recommended beforehand!

84

84

Not all shifts are equal

- Sudden shifts vs. gradual shifts
 - Sudden shifts are easier to detect than gradual shifts



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85

85

Not all shifts are equal

- Sudden shifts vs. gradual shifts
- Spatial shifts vs. temporal shifts

- New device (e.g. mobile vs. desktop)
- New users (e.g. new country)

E.g. same users, same device, but behaviors change over time

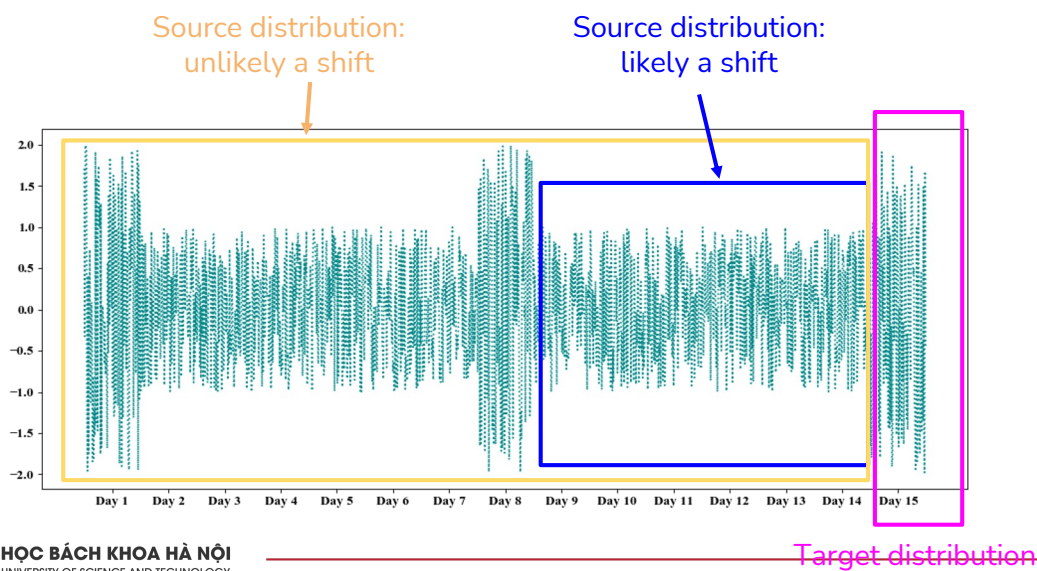


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86

86

Temporal shifts: time window scale matters

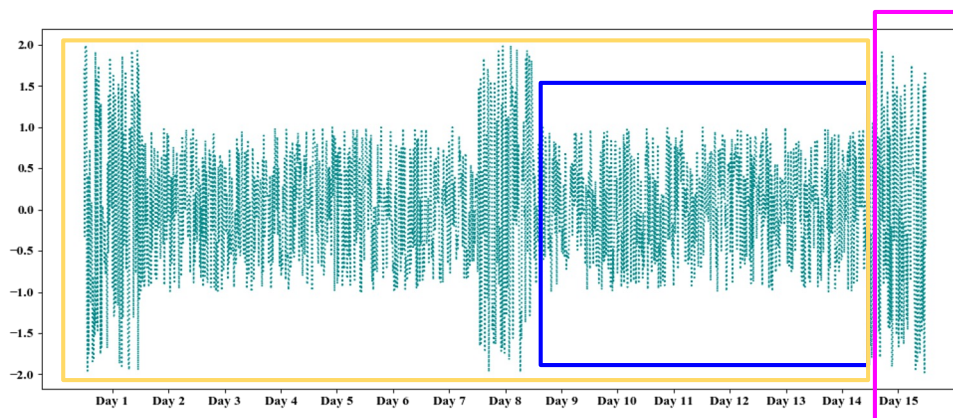


87

87

Temporal shifts: time window scale matters

Difficulty is compounded
by seasonal variation



88

88

Temporal shifts: time window scale matters

- Too short window: false alarms of shifts
- Too long window: takes long to detect shifts

- Granularity level: hourly, daily



Temporal shifts: time window scale matters

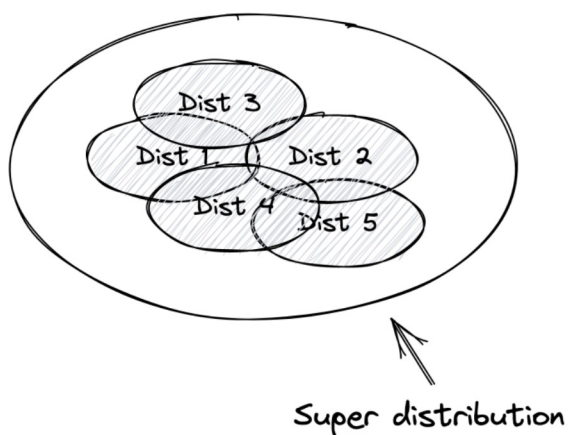
- Too short window: false alarms of shifts
- Too long window: takes long to detect shifts

- Granularity level: hourly, daily
- Merge shorter time scale windows -> larger time scale window
- RCA: automatically analyze various window sizes



Addressing data distribution shifts

1. Train model using a massive dataset



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91

Addressing data distribution shifts

1. Train model using a massive dataset
2. Retrain model with new data from new distribution
 - Mode
 - Train from scratch
 - Fine-tune



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92

92

Addressing data distribution shifts

1. Train model using a massive dataset
2. Retrain model with new data from new distribution
 - Mode
 - Data
 - Use data from when data started to shift
 - Use data from the last X days/weeks/months
 - Use data from the last fine-tuning point

Need to figure out not just when to retrain models, but also how and what data



Monitoring & Observability



Monitoring vs. observability

- Monitoring: tracking, measuring, and logging different metrics that can help us **determine when something goes wrong**
- Observability: setting up our system in a way that gives us visibility into our system to **investigate what went wrong**



Monitoring vs. observability

- Monitoring: tracking, measuring, and logging different metrics that can help us **determine when something goes wrong**
- Observability: **setting up our system** in a way that gives us visibility into our system to **investigate what went wrong**

Instrumentation

- adding timers to your functions
- counting NaNs in your features
- logging unusual events e.g. very long inputs
- ...



Monitoring vs. observability

- Monitoring: tracking, measuring, and logging different metrics that can help us **determine when something goes wrong**
- Observability: setting up our system in a way that gives us visibility into our system to **investigate what went wrong**

Observability is part of monitoring



Monitoring is all about metrics

- Operational metrics
- ML-specific metrics



Operational metrics

- Latency
- Throughput
- Requests / minute/hour/day
- % requests that return with a 2XX code
- CPU/GPU utilization
- Memory utilization
- Availability
- etc.



Operational metrics

- Latency
- Throughput
- Requests / minute/hour/day
- % requests that return with a 2XX code
- CPU/GPU utilization
- Memory utilization
- Availability
- etc.

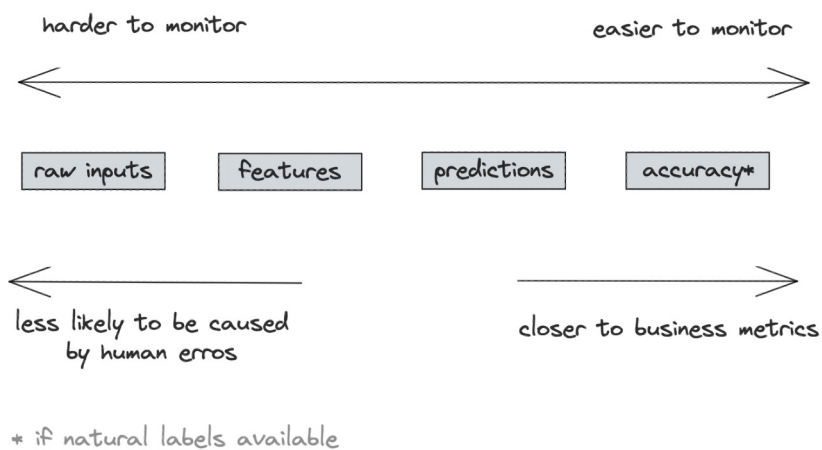
SLA example

- Up means:
 - median latency <200ms
 - 99th percentile <2s
- 99.99% uptime (four-nines)

SLA for ML?



ML metrics: what to monitor



Monitoring #1: accuracy-related metrics

- Most direct way to monitor a model's performance
 - Can only do as fast as when feedback is available



Monitoring #1: accuracy-related metrics

- Most direct way to monitor a model's performance
- Collect as much feedback as possible
- Example: YouTube video recommendations
 - Click through rate
 - Duration watched
 - Completion rate
 - Take rate



Monitoring #2: predictions

- Predictions are low-dim: easy to visualize, compute stats, and do two-sample tests
- Changes in prediction dist. generally mean changes in input dist.



Monitoring #2: predictions

- Predictions are low-dim: easy to visualize, compute stats, and do two-sample tests
- Changes in prediction dist. generally mean changes in input dist.
- Monitor odd things in predictions
 - E.g. if predictions are all False in the last 10 mins



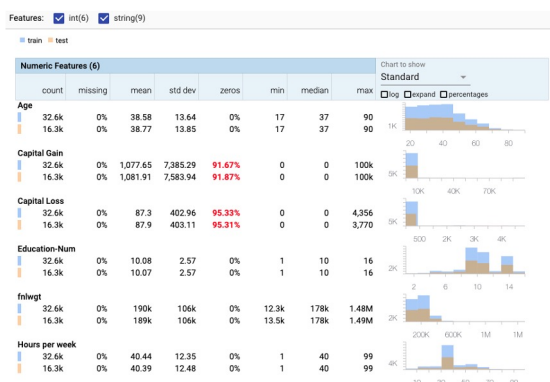
Monitoring #3: features

- Most monitoring tools focus on monitoring features
- Feature schema expectations
 - Generated from the source distribution
 - If violated in production, possibly something is wrong
- Example expectations
 - Common sense: e.g. “the” is most common word in English
 - min, max, or median values of a feature are in [a, b]
 - All values of a feature satisfy a regex
 - Categorical data belongs to a predefined set
 - `FEATURE_1 > FEATURE_B`



Generate expectations with profiling & visualization

- Examining data & collecting:
 - statistics
 - informative summaries
- [pandas-profiling](#)
- [facets](#)



Report generated with pandas-profiling

Out[5]:

```
In [ ]: M # Or use the HTML report in an iframe
        profile.to_notebook_iframe()
```

Age	Cabin	Embarked	Fare	Name	Parch	PassengerId	Pclass	Sex	Smoker	Survived	Ticket
33.0	NaN	S	7.8958	Markun, Mr. Johann	0	882	3	male	0	0	348257
32.0	NaN	S	10.5167	Dahlberg, Mrs. Gerta Ulrika	0	883	3	female	0	0	7952
28.0	NaN	S	10.5000	Bartlett, Mr. Frederick James	0	884	2	male	0	0	C.A. 830703 34058
25.0	NaN	S	7.0500	Sulekall, Mr. Henry Jr	0	885	3	male	0	0	8070403 362076
39.0	NaN	O	29.1250	Rice, Mrs. William (Margaret Horton)	5	886	3	female	0	0	362652
27.0	NaN	S	13.0000	Montvila, Rev. Juozas	0	887	2	male	0	0	211536
19.0	842	S	30.0000	Graham, Mrs. Margaret Edith	0	888	1	female	0	1	112053
NaN	NaN	S	23.4500	Johnston, Mrs. Catherine Helen "Carnie"	2	889	3	female	1	0	W.C. 6607
26.0	C148	C	30.0000	Behr, Mr. Karl Howell	0	890	1	male	0	1	111369
32.0	NaN	O	7.7500	Doolley, Mr. Patrick	0	891	3	male	0	0	370376

107

107

Monitoring #3: features

- Feature schema expectations

Table shape

- `expect_column_to_exist`
- `expect_table_columns_to_match_ordered_list`
- `expect_table_columns_to_match_set`
- `expect_table_row_count_to_be_between`
- `expect_table_row_count_to_equal`
- `expect_table_row_count_to_equal_other_table`

Missing values, unique values, and types

- `expect_column_values_to_be_unique`
- `expect_column_values_to_not_be_null`
- `expect_column_values_to_be_null`
- `expect_column_values_to_be_of_type`
- `expect_column_values_to_be_in_type_list`

```
expect_column_values_to_be_between(
    column="room_temp",
    min_value=60,
    max_value=75,
    mostly=.95
)
```

"Values in this column should be between 60 and 75, at least 95% of the time."

"Warning: more than 5% of values fell outside the specified range of 60 to 75."



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[GitHub - great-expectations/great_expectations](#)

108

Monitoring #3: features schema with pydantic

```
from pydantic import BaseModel, ValidationError, validator
```

```
class UserModel(BaseModel):
    name: str
    username: str
    password1: str
    password2: str

    @validator('name')
    def name_must_contain_space(cls, v):
        if ' ' not in v:
            raise ValueError('must contain a space')
        return v.title()

    @validator('password2')
    def passwords_match(cls, v, values, **kwargs):
        if 'password1' in values and v != values['password1']:
            raise ValueError('passwords do not match')
        return v

    @validator('username')
    def username_alphanumeric(cls, v):
        assert v.isalnum(), 'must be alphanumeric'
        return v
```

```
user = UserModel(
    name='samuel colvin',
    username='scolvin',
    password1='zxcvbn',
    password2='zxcvbn',
)
print(user)
#> name='Samuel Colvin' username='scolvin' password1='zxcvbn' password2='zxcvbn'

try:
    UserModel(
        name='samuel',
        username='scolvin',
        password1='zxcvbn',
        password2='zxcvbn2',
    )
except ValidationError as e:
    print(e)
    """
2 validation errors for UserModel
name
  must contain a space (type=value_error)
password2
  passwords do not match (type=value_error)
    """
```



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<https://pydantic-docs.helpmanual.io/usage/validators/>

109

109

Monitoring #3: features schema with TFX

```
# Generate training stats & schema
train_stats = tfdv.generate_statistics_from_dataframe(df)
schema = tfdv.infer_schema(statistics=train_stats)
```

schema

```
feature {
  name: "l"
  type: FLOAT
  presence {
    min_fraction: 1.0
    min_count: 1
  }
  shape {
    dim {
      size: 1
    }
  }
}
```

```
# Generate serving stats
serving_stats = tfdv.generate_statistics_from_dataframe(serving_df)
# Domain knowledge required
tfdv.get_feature(schema, "diabetesMed").skew_comparator.infinity_norm.threshold = 0.03
# Compare serving stats to training stats to detect skew
skew_anomalies = tfdv.validate_statistics(
    statistics=train_stats,
    schema=schema,
    serving_statistics=serving_stats)
```

Anomaly short description		Anomaly long description
Feature name		
'payer_code'	High Linfty distance between current and previous	The Linfty distance between current and previous is 0.0342144 (up to six significant digits), above the threshold 0.03. The feature value with maximum difference is: MC
'diabetesMed'	High Linfty distance between training and serving	The Linfty distance between training and serving is 0.0325464 (up to six significant digits), above the threshold 0.03. The feature value with maximum difference is: No



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[How To Evaluate MLOps Tools](#) (Hamel Husain, CS 329S Lecture 9, 2022)

110

110

Feature monitoring problems

1. Compute & memory cost
 - a. 100s models, each with 100s features
 - b. Computing stats for 10000s of features is costly



111

Feature monitoring problems

1. Compute & memory cost
2. Alert fatigue
 - a. Most expectation violations are benign



112

Feature monitoring problems

1. Compute & memory cost
2. Alert fatigue
3. Schema management
 - a. Feature schema changes over time
 - b. Need to find a way to map feature to schema version



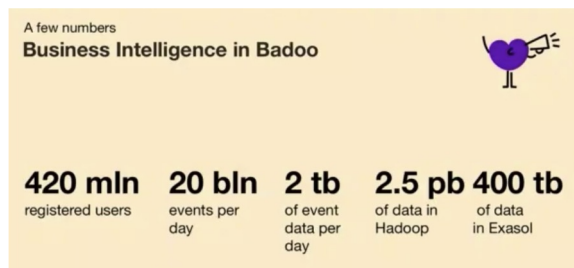
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113

113

Monitoring toolbox: logs

- Log everything
- A stream processing problem



Vladimir Kazanov ([Badoo 2019](#))

"If it moves, we track it. Sometimes we'll draw a graph of something that isn't moving yet, just in case it decides to make a run for it."

Ian Malpass ([Etsy 2011](#))



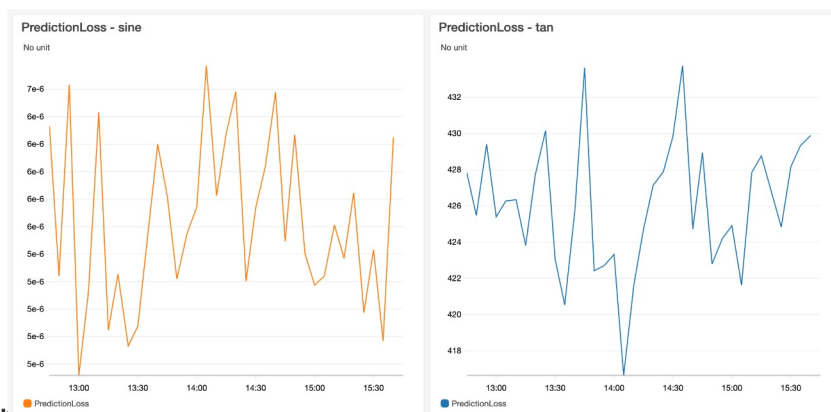
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114

114

Monitoring toolbox: dashboards

- Make monitoring accessible to non-engineering stakeholders
- Good for visualizations but insufficient for discovering distribution shifts



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115

115

Monitoring toolbox: alerts

- 3 components
 - Alert policy: condition for alert
 - Notification channels
 - Description
- Alert fatigue
 - How to send only meaningful alerts?

Recommender model accuracy below 90%

\${timestamp}: This alert originated from the service \${service-name}



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116

116

Monitoring -> Continual Learning

- Monitoring is passive
 - Wait for a shift to happen to detect it
- Continual learning is active
 - Update your models to address shifts

